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Original article

Association between male pattern baldness and prostate disease: A meta-analysis

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Abstract

Background: Male pattern baldness (MPB) has been associated with an increased risk of prostate cancer (PC) as well as benign prostatic hyperplasia (BPH). We performed a meta-analysis to quantitatively determine the level of risk of PC and BPH in individuals with baldness.

Methods: A systematic literature search was conducted using several databases. We calculated pooled odds ratios (OR) and 95% CIs.

Results: In total, 17 studies comprising 68,448 participants were eligible for the meta-analysis and showed that MPB is associated with an increased risk of aggressive PC (OR = 1.59; 95% CI: 1.36–1.86; $P < 0.001$) as well as BPH (OR = 1.26; 95% CI: 1.05–1.51; $P = 0.01$). There was statistically significant association between vertex baldness and PC (OR = 1.18; 95% CI: 1.05–1.32; $P = 0.006$). No statistically significant association between vertex, frontal plus vertex hair loss pattern, and BPH were identified.

Conclusions: MPB is associated with an increased risk of PC and BPH. Despite our findings, further studies, preferably prospective cohort studies, are required to better elucidate these relationships and to advance knowledge in this field. © 2018 Elsevier Inc. All rights reserved.

Keywords: Baldness; Androgenetic alopecia; Prostate cancer; Benign prostatic hyperplasia

1. Introduction

Male pattern baldness (MPB) is the most common cause of hair loss mediated by systemic androgens and genetic factors, which affects up to 70% of men and increases with age [1]. Androgens and the androgen receptor are required for expression of the male phenotype. However, MPB is an androgen-independent genetic disorder which is induced via the activation of androgen receptor in hair follicles by dihydrotestosterone. Inhibition of dihydrotestosterone production can suppress the progression of MPB which can explain the phenomenon that eunuchs do not develop baldness if castrated before the age of 25 [2]. Androgens have also been strongly implicated in the carcinogenesis of prostate cancer (PC) and the development and maintenance of benign prostatic hyperplasia (BPH) [3,4]. In addition, previous studies identified that finasteride, a 5α-reductase inhibitor, is the main agents for MPB and BPH because of they respond to the inhibition of 5α-reductase. Therefore, MPB seems to share similarly pathologic mechanisms with PC and BPH for aging and androgen influence.

PC is one of the most commonly diagnosed cancers and one of the most common causes of cancer related death in developed countries. Although some risk factors have been established for PC, they collectively explain only a fraction of the disease occurrence. Therefore, additional research to improve our understanding of the etiology of PC is needed. Several studies have investigated the relationship between MPB and PC and have yielded inconsistent results [5–9].

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Recently, a meta-analysis [10] reported vertex baldness (but not frontal baldness) to be associated with an increased risk of PC, however, it only included 7 studies in its analysis and did not consider the relationship between MPB and BPH. Therefore, the aim of this study is to conduct a comprehensive meta-analysis of existing studies to confirm the association of MPB with PC and BPH.

2. Methods

This study protocol was approved by the institutional review board at the West China School of Medicine before initiation and it did not need ethical standard statement.

2.1. Data sources and searches

A systematic literature search in the MEDLINE, EMBASE, and Cochrane databases was conducted to identify relevant studies which reported the association of MPB with the risk of PC and BPH published up to January 2017. Additional manual searches were made using the reference lists from relevant studies to retrieve other papers relevant to our topic.

2.2. Data extraction and study quality

Two reviewers reviewed the full texts of the included studies which met the following inclusion criteria were included in the meta-analysis: (1) The risk point estimate was reported as an odds ratio (OR) with the 95% CI, or the data were presented such that a OR and 95% CI could be calculated; (2) the study had a cohort design or retrospective case-control design or randomized control design; (3) the study evaluated MPB with PC or BPH; and (4) the study language was published in English. Studies that did not meet the earlier criteria were excluded. Data were extracted via a standardized data extraction form, collecting information on the year of publication, country, study design, number of cases and controls, type of alopecia, severity of alopecia, and PC stage, when available. The methodological quality of included studies was assessed by 2 authors using the Downs and Black tool for both randomized controlled trials and nonrandomized controlled trial [11]. In general, the score ranges were grouped into the following 4 quality levels: excellent (26–28), good (20–25), fair (15–19), and poor (<14).

2.3. Statistical analysis and meta-analysis

Heterogeneity, that is, the proportion of variability across studies was assessed using Cochran's Q statistic and quantified using the I^2 statistic. If high heterogeneity ($I^2 > 50$) was found, influence analysis (one kind method of sensitivity analysis) would be performed. We would take out included studies one by one ("leave one out") to see which study influenced the heterogeneity of the meta-analysis. Pooled estimates were calculated omitting one study at a time. For all statistical analyses, a 2-sided $P < 0.05$ was considered statistically significant. Data analysis were performed with Review Manager Software (RevMan v.5.2, Cochrane Collaboration, Oxford, UK).

3. Results

3.1. Characteristics of studies

A Preferred Reporting Items of Systematic Reviews and meta-analyses (PRISMA) [12] flowchart of screening and selection results shown in Fig. 1. Using our prespecified search strategy, 203 extracts were retrieved and 7 additional citations were obtained through reference lists. From 210 studies initially identified, 25 were considered potentially suitable. After a full-text review, 17 studies [5–9,13–24] with 68,448 participants met inclusion criteria and were included in the final analysis (Fig. 1). General study characteristics were presented in Table 1. Of the 17 studies, 12 was about PC [5–9,13–18,24] (2 were cohort studies), 6 was about BPH [15,19–23] (1 reported both PC and BPH [15]). Seven studies included in this meta-analysis evaluated baldness patterns at different ages [5–9,13,16]. Table 2 shows the association between different types of MPB and PC or BPH in every included study.

3.2. Association between MPB and PC

We examined the relationship between any pattern baldness and the risk of PC using data from 11 studies. There was no significant association between any pattern baldness and PC risk (OR = 1.05; 95% CI: 0.98–1.12; $P = 0.19$) (Fig. 2A). Meta-analysis of 5 case-control studies revealed statistically significant association between any pattern baldness and aggressive PC (OR = 1.60; 95% CI: 1.37–1.86; $P < 0.001$; $I^2 = 0\%$) (Fig. 2B). The later inclusion of the study by Zhou et al. [18] as the "leave one out" sensitivity analysis (influence analysis), the pooled results did not change substantially (OR = 1.48; 95% CI: 1.18–1.85; $P = 0.0006$), however, it introduced significant heterogeneity into the analysis ($I^2 = 65\%$) (Supplemental Fig. 1). Therefore, we excluded it from the analysis. There were 5 case-control studies evaluated the association between baldness and nonaggressive PC. We found no statistically significant association between them (OR = 1.04; 95% CI: 0.92–1.19; $P = 0.52$; $I^2 = 18\%$) (Fig. 2C).

3.3. Frontal baldness and risk of PC

Meta-analysis of 10 studies revealed no statistically significant association between frontal baldness and PC (OR = 0.98; 95% CI: 0.90–1.06; $P = 0.56$; $I^2 = 23\%$) (Fig. 3A). The later inclusion of the study by Thomas et al.

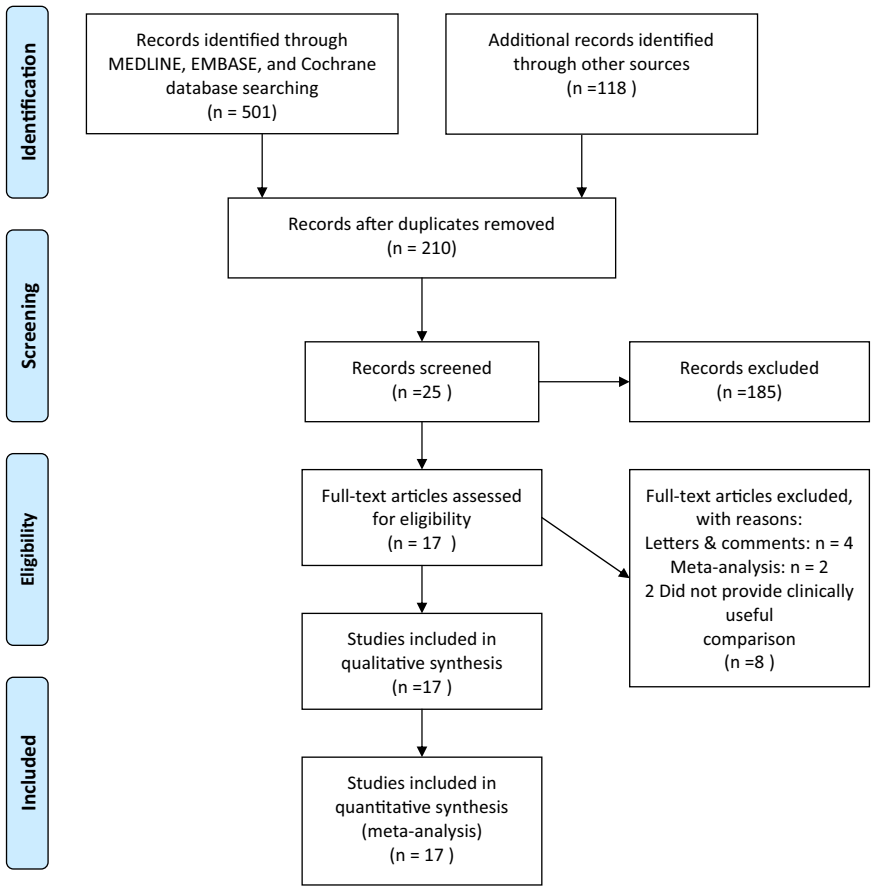

Fig. 1. Flowchart for records selection process of the meta-analysis (according to PRISMA template: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. http://dx.doi.org/10.1371/journal.pmed1000097).

[7] as a sensitivity analysis, not affected the result (pooled OR = 1.04; 95% CI: 0.95–1.13; $P = 0.38$), however, it also introduced significant heterogeneity into the analysis ($I^2 = 83\%$). One possible reason was that the study had small number of participants and included only veterans. Meta-analysis of 3 case-control studies revealed a statistically significant association between frontal baldness and aggressive PC (OR = 1.27; 95% CI: 1.03–1.55; $P = 0.02$;

Table 1
Summary of studies included in the meta-analysis

Study	Country	Study type	Quality score	Cases	Controls/participants	Outcome
Yassa et al. [5]	France	Case-control	18	388	281	PCa
Thomas et al. [7]	USA	Case-control	17	312	167	PCa
Cremers et al. [8]	Netherlands	Case-control	18	938	1,911	PCa
Wright et al. [9]	USA	Case-control	18	999	942	PCa
Zeigler-Johnson et al. [13]	USA	Case-control	19	318	219	PCa
Giles et al. [14]	Australia	Case-control	16	1446	1,390	PCa
Faydaci et al. [15]	Turkey	Case-control	17	44	108	PCa
Demark-Wahnefried et al. [16]	USA	Case-control	18	134	145	PCa
Demark-Wahnefried et al. [17]	USA	Case-control	17	129	139	PCa
Muller et al. [6]	Australia	cohort study	20	476	9,448	PCa
Zhou et al. [18]	USA	cohort study	20	1138	39,070	PCa
Sarre et al. [24]	Finnish	cohort study	19	757	11,795	PCa
Chen et al. [19]	Taiwan	Case-control	17	42	34	BPH
Oh et al. [20]	South Korea	Case-control	16	225	160	BPH
Barbosa et al. [21]	Latin American	cohort study	20	410	907	BPH
Fritschi et al. [22]	Australia	Case-control	18	398	471	BPH
Severi et al. [23]	Australia	Case-control	19	129	1,261	BPH
Faydaci et al. [15]	Turkey	Case-control	19	108	44	BPH

Table 2
Association between different types of male pattern boldness (MPB) and prostate cancer (PC) or benign prostate hyperplasia (BPH)

Study	Types of MPB	MPB with PC		MPB with aggressive PC		MPB with nonaggressive PC		MPB with BPH	
		OR/HR	95% CI	OR/HR	95% CI	OR/HR	95% CI	OR/HR	95% CI
Yassa et al. [5]	Any	0.96	0.7–1.3	1.74	1.17–2.59	0.91	0.64–1.31	N/A	
	Frontal	1.01	0.69–1.48						
	Vertex	0.92	0.6–1.41						
	Frontal plus vertex	0.9	0.52–1.57						
Thomas et al. [7]	Any	1.78	1.2–2.66	2.38	1.34–4.23	1.14	0.73–1.79	N/A	
	Frontal	4.62	2.99–7.14						
	Vertex								
	Frontal plus vertex								
Cremers et al. [8]	Any	1.17	0.96–1.42	1.49	1.18–1.88	1.03	0.77–1.36	N/A	
	Frontal	1.1	0.88–1.38	1.12	0.86–1.47	1.07	0.77–1.49		
	Vertex	1.28	1.01–1.61	1.37	1.04–1.8	1.14	0.81–1.6		
	Frontal plus vertex	1.13	0.9–1.43	1.3	0.99–1.71	0.86	0.61–1.24		
Wright et al. [9]	Any	0.87	0.72–1.04					N/A	
	Frontal	0.76	0.61–0.95						
	Vertex								
	Frontal plus vertex								
Zeigler-Johnson et al. [13]	Any	1.65	1.02–2.66	1.44	0.79–2.64	1.79	1.05–3.03	N/A	
	Frontal	1.85	0.98–3.46	2.15	1.03–4.49	1.59	0.78–3.27		
	Vertex	1.44	0.74–2.83	0.68	0.24–1.94	0.89	0.44–1.80		
	Frontal plus vertex			0.68	0.24–1.94	1.99	0.98–4.05		
Giles et al. [14]	Any	1.11	0.93–1.32	1.55	1.15–2.08	1	0.84–1.21	N/A	
	Frontal	1.02	0.83–1.24	1.37	0.98–1.92	0.94	0.76–1.16		
	Vertex	1.35	1.08–1.7	2	1.39–2.87	1.2	0.94–1.53		
	Frontal plus vertex	1.06	0.86–1.3	1.46	1.03–2.08	0.96	0.77–1.2		
Faydaci et al. [15]	Any	0.94	0.34–2.64					1.06	0.38–2.96
	Frontal	0.78	0.22–2.71						
	Vertex	1	0.35–2.85						
	Frontal plus vertex								
Demark-Wahnefried et al. [16]	Any	1.17	0.72–1.9					N/A	
	Frontal	0.82	0.41–1.63						
	Vertex	1.9	1.01–3.58						
	Frontal plus vertex	0.58	0.19–1.76						
Demark-Wahnefried et al. [17]	Any	0.88	0.54–1.43					N/A	
	Frontal	0.63	0.26–1.49						
	Vertex	0.94	0.57–1.56						
	Frontal plus vertex								
Muller et al. [6]	Any	0.96	0.79–1.16					N/A	
	Frontal	0.92	0.69–1.22						
	Vertex	0.98	0.79–1.23						
	Frontal plus vertex								
Zhou et al. [18]	Any	1.03	0.92–1.16	1.06	0.90–1.25	0.97	0.82–1.15	N/A	
	Frontal	1	0.87–1.16	1	0.81–1.23	1	0.82–1.24		
	Vertex								
	Frontal plus vertex	1	0.87–1.16	1.11	0.92–1.35	0.95	0.77–1.16		
Sarre et al. [24]	Any							N/A	
	Frontal or vertex	0.91	0.78–1.06						
	Almost or completely bald	0.81	0.56–1.18						
Barbosa et al. [21]	Any	N/A						1.36	1.00–1.84
Chen et al. [19]	Any	N/A						2.5	0.80–7.79
Fritschi et al. [22]	Any	N/A						1.06	0.81–1.39
	Vertex							1.07	0.80–1.45
	Frontal plus vertex							1.04	0.67–1.61
Oh et al. [20]	Any	N/A						2.96	1.88–4.65
Severi et al. [23]	Any	N/A						1.55	0.99–2.42
	Vertex							1.46	0.82–2.60
	Frontal plus vertex							1.78	1.07–2.96

HR = hazard ratio; N/A = not applicable.

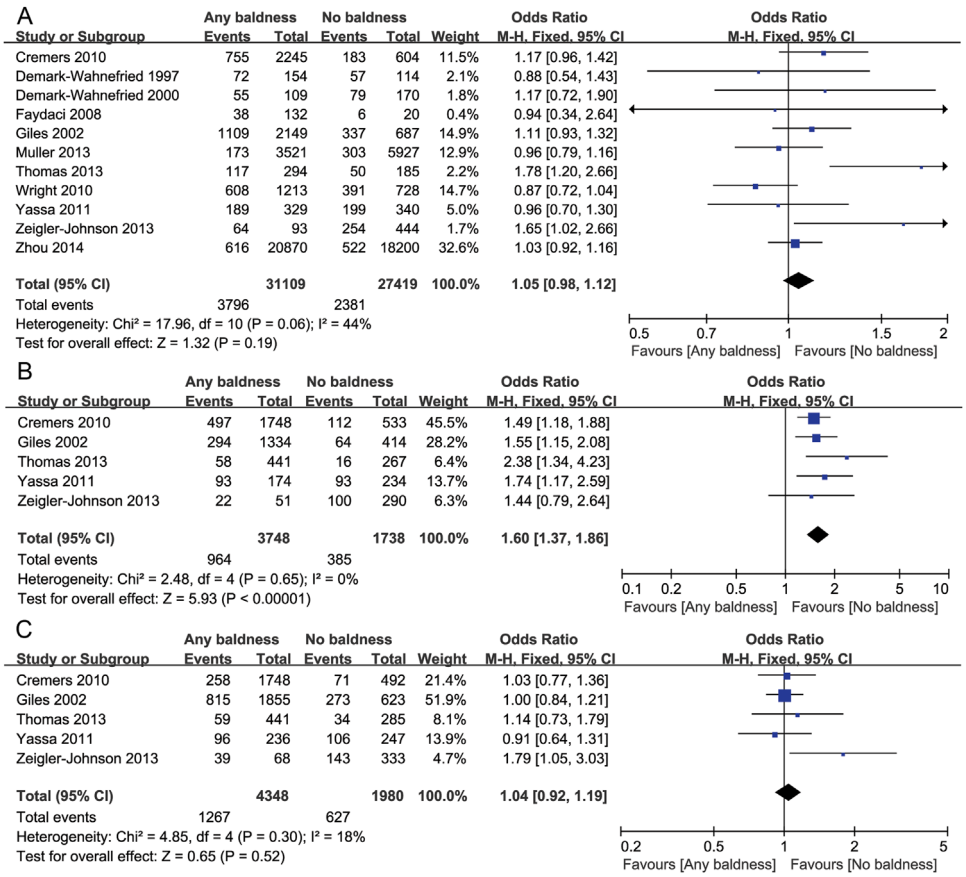

Fig. 2. Association between male pattern baldness and the risk of prostate cancer. (A) Any pattern baldness and prostate cancer. (B) Any pattern baldness and aggressive prostate cancer. (C) Any pattern baldness and nonaggressive prostate cancer.

I^2 = 33%) (Fig. 2B). However, there was no significant association between frontal baldness and nonaggressive PC risk (OR = 1.00; 95% CI: 0.88–1.15; P = 0.96; I^2 = 0%) (Fig. 3C).

3.4. Vertex baldness and PC

Three studies did not classify the MPB into vertex baldness. Meta-analysis of 8 other studies revealed statistically significant association between vertex baldness and PC (pooled OR = 1.18; 95% CI: 1.05–0.32; P = 0.006; I^2 = 24%) (Fig. 4A). There were 2 case-control studies evaluated the association between vertex baldness and aggressive PC. We found a statistically significant association between them (OR = 1.62; 95% CI: 1.12–2.35; P = 0.01) (Fig. 4B). There was no statistically significant heterogeneity (I^2 = 63%, P = 0.10). However, there was no significant association between vertex baldness and nonaggressive PC risk (OR = 1.18; 95% CI: 0.97–1.44; P = 0.10; I^2 = 0%) (Fig. 4C).

3.5. Frontal plus vertex baldness and PC

We examined the relationship between frontal plus vertex baldness and the risk of PC using data from 6 studies. There was no significant association between frontal plus vertex baldness and PC risk (OR = 1.02; 95% CI: 0.93–1.12; P = 0.71; I^2 = 0%) (Fig. 5A). The meta-analysis of 3 case-control studies showed an increased risk of aggressive PC for patients with frontal plus vertex baldness (OR = 1.32; 95% CI: 1.07–1.63; P = 0.009; I^2 = 0%) (Fig. 5B). There was no association between frontal plus vertex baldness and nonaggressive PC in case-control studies (OR = 1.04; 95% CI: 0.75–1.43; P = 0.83) (Fig. 5C). There was nonstatistically significant heterogeneity (I^2 = 54%, P = 0.11).

3.6. Association between MPB and BPH

Meta-analysis of 5 studies revealed a statistically significant association between MPB and BPH (OR = 1.26; 95% CI: 1.06–1.51; P = 0.01; I^2 = 1%) (Fig. 6). The later inclusion of the study by Oh et al. [20] as a sensitivity analysis, not affected the result (OR = 1.56; 95% CI: 1.10–2.19; P = 0.01), however, it also introduced significant heterogeneity into the analysis (I^2 = 69%, P = 0.007). Four studies did not classify the MPB into frontal and vertex baldness. Meta-analysis of 2 studies [22,23] revealed no significant association between vertex, frontal plus vertex hair loss pattern and BPH.

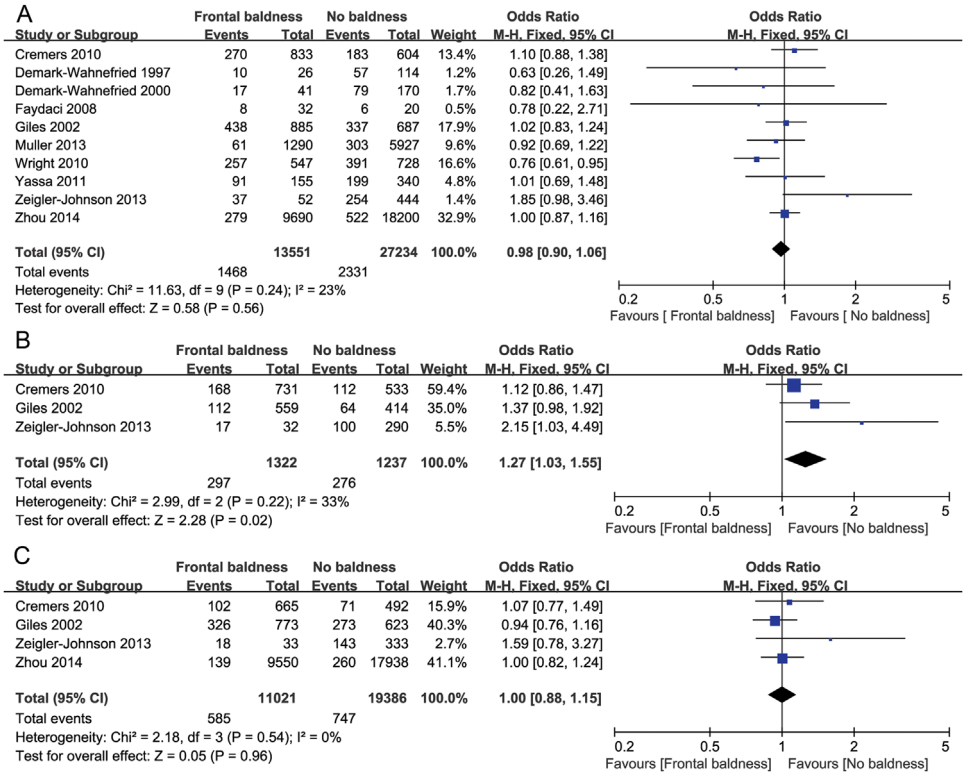

Fig. 3. Association between frontal baldness and the risk of prostate cancer. (A) Frontal baldness and prostate cancer. (B) Frontal baldness and aggressive prostate cancer. (C) Frontal baldness and nonaggressive prostate cancer.

4. Discussion

The main finding of our meta-analysis is that baldness is positively associated with an increased risk of aggressive PC as well as BPH, compared to those without baldness. Our study is the most comprehensive meta-analysis to date that examines the association between MPB and prostate disease with 68,448 participants. It is the first meta-analysis

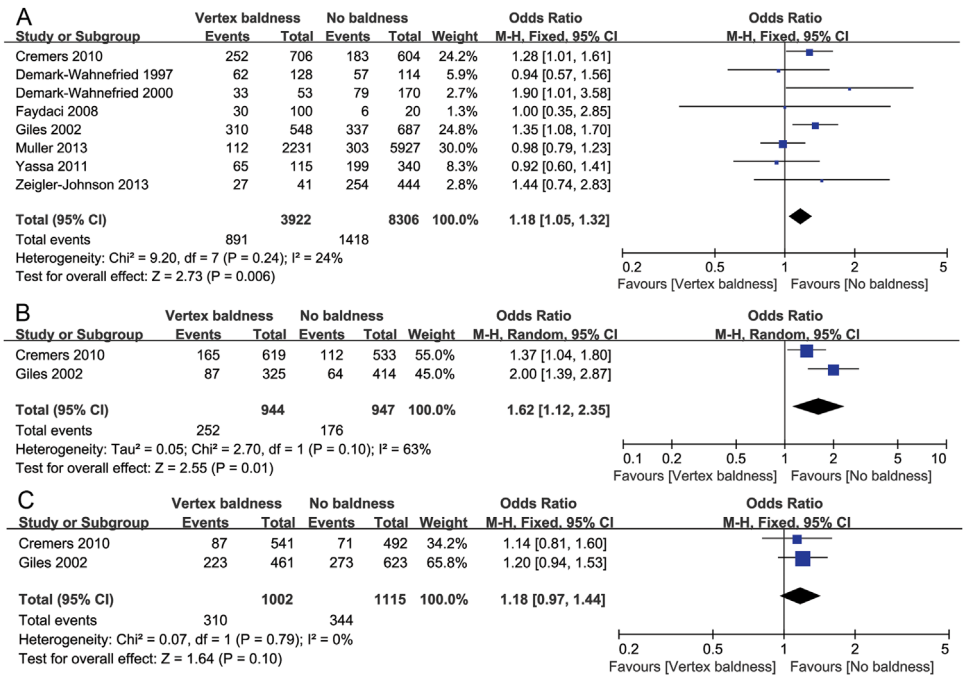

Fig. 4. Association between vertex baldness and the risk of prostate cancer. (A) Vertex baldness and prostate cancer. (B) Vertex baldness and aggressive prostate cancer. (C) Vertex baldness and nonaggressive prostate cancer.

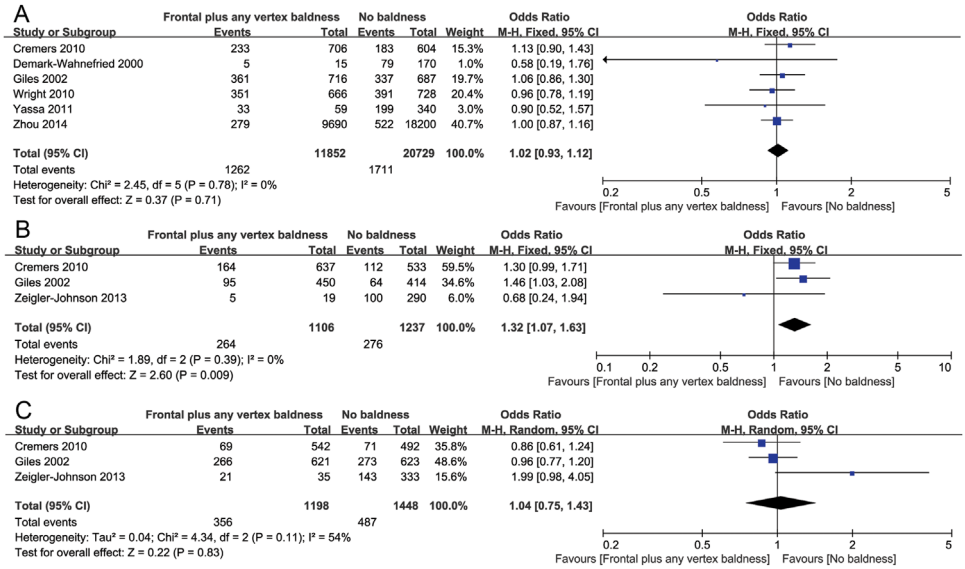

Fig. 5. Association between frontal plus vertex baldness and the risk of prostate cancer. (A) Frontal plus vertex baldness and prostate cancer. (B) Frontal plus vertex baldness and aggressive prostate cancer. (C) Frontal plus vertex baldness and nonaggressive prostate cancer.

to examine the relationship between MPB and aggressive PC and BPH. In the literature, it has been one previous meta-analysis on the effect of androgenetic alopecia on PC [10]. That meta-analysis assessed 7 case-control studies (n = 8,994) and reported that vertex baldness was associated with an increased risk of PC but that frontal baldness was not. However, that study had some limitations, including that it only considered the relationship between alopecia and PC, subtype-specific analyses by PC aggressiveness were not presented. Our meta-analysis overcomes these limitations by incorporating 17 studies and by including analyses of the relationships between MPB, aggressive PC, and BPH.

The pathological mechanisms underlying the relationship between MPB and prostate disease have yet to be clearly established. The leading hypothesis is that androgens play an extremely important role in the pathophysiology of both PC and baldness, given that androgens profoundly affect the development of PC as well as the physiological growth of scalp and body hair. A lot of evidence from basic science and observational studies supports the hypothesis. Eunuchs rarely diagnosed with PC and did not go bald. Castration has a palliative effect on PC and castrated men also do not go bald [25]. Testosterone alone can produce prostatic adenocarcinoma [25]. Bald men have been shown to have a greater number of androgen receptors in the scalp and higher serum levels of testosterone [17]. As well as men with androgen insensitivity syndrome or testicular feminization, in whom functional androgen receptors are lacking, also do not go bald [25]. In addition to androgens, there are some other similarities between the 2 disorders such as aging and hereditability. PC is age-dependent disease, which its incidence increases that come with age, as well as going bald.

BPH is a disease that impairs the quality of life in many aging men. Previous studies have suggested an association between MPB and BPH with respect to aging and endogenous hormones. Epidemiologic associations between MPB and BPH have been studied in face of a possible common underlying mechanism for both conditions. MPB prevalence increases with age, as well as BPH which is age-related prevalence of 50% for men in the 51 to 60 years age group and up to 70% in the 61 to 70 years age group. Like MPB, BPH is also androgen-dependent entity, which androgens play a permissive role in the development of prostatic growth [4]. Previous evidence showed that individuals with innate 5α-reductase deficiency did not go MPB or BPH during their lifetime [26]. These findings may underlie a basic mechanism that could potentially explain the results we observed in present study. However, further

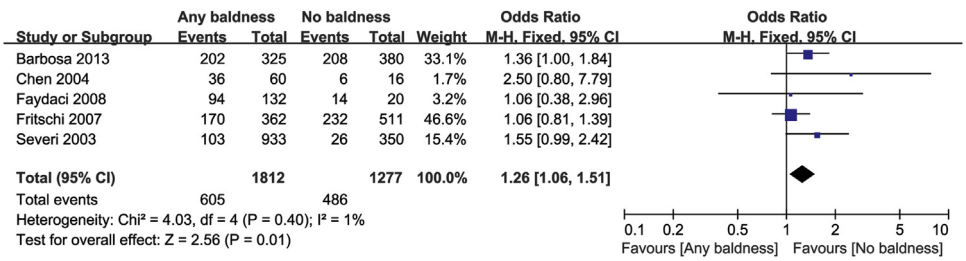

Fig. 6. Association between male pattern baldness and the risk of benign prostatic hyperplasia.

researches are necessary to better understand the potential association between PC, BPH, and MPB.

Although the association between MPB and PC has been inconsistent, recent cohort studies suggested a positive relationship [6,10,18]. Our findings suggest that bald men at any age is significantly associated with aggressive PC risk relative no baldness, regardless of the classification (any, frontal, vertex, and frontal plus vertex baldness). One such cohort study [18] comprised 39,070 men, 1,138 of whom were diagnosed with PC during 2.78 years of follow-up. This study predicted a 1.19 times higher risk of aggressive PC among men with frontal plus moderate vertex baldness at age 45 years than those without baldness. Another cohort study, which assessed androgenetic alopecia at ages 20 and 40 years, found a 1.81 times higher risk of PC by age 55 years among men with vertex baldness at age 40 years than those without baldness. In addition, recent case-control study found frontal baldness was associated with high-stage (OR = 2.61, 95% CI: 1.10–6.18) and high-grade (OR = 2.20, 95% CI: 1.05–4.61) PC. The earlier results are consistent with our findings. However, a number of studies had found no association between MPB and PC [15,17], others had found MPB to be associated with a decreased risk of PC [8,9]. Although differences in sample size, study design, and exposure assessment might plausibly explain much of the inconsistency across studies. However, there was not uniformity in the specific ages studied and exist data heterogeneity precluded dose-respond meta-analysis. In a smaller case-control study, investigating a possible relation between early-onset vertex baldness at ages 30 and 40 years and PC, early frontal baldness was associated with a decreased risk of PC (at age 30 years: OR = 0.66, 95% CI: 0.19–2.35; at age 40 years: OR = 0.88, 95% CI: 0.40-1.94). Muller et al. [6] finished the cohort study found that the hazard of PC is 1.81 (95% CI: 1.13–2.90) times higher for those with vertex baldness than those with no baldness at age 55 years, however, the vertex baldness had a 44% lower hazard than those with no baldness at age 75 years. Other case-control study [13] observed significant associations between early-onset baldness and PC in African-American men. Found PC diagnosed less than the age of 60 years, frontal baldness was associated with high stage (OR = 6.51, 95% CI: 2.11–20.06) and high grade (OR = 4.23, 95% CI: 1.47–12.14). No significant associations between baldness and PC severity were observed for older men (age > 60 years). With regard to the age-dependent association between MPB and risk of PC, further studies are warranted.

The association between MPB phenotype (vertex, frontal, or vertex plus frontal baldness) and PC was also examined. In the 17 included studies, they all evaluated the associations between different MPB and PC. For example, in Zhou et al's research [18], they observed a positive and statistically significant association between frontal with moderate vertex baldness and aggressive PC (hazard ratio¼ = 1.39, 95% CI¼: 1.07–1.80). In the study of Thomas et al. [7], they found that the increased overall risk was more strongly linked to frontal baldness than vertex baldness as reported by prior studies. Our pool results of association between different MPB phenotype and PC were diverse. Meta-analysis of 10 studies revealed no statistically significant association between frontal baldness and PC. However, meta-analysis of 8 other studies revealed statistically significant association between vertex baldness and PC. For frontal plus vertex baldness, there was no significant association between it and PC risk. The pathophysiological differences between frontal and vertex baldness are unknown. In the future, as the pathophysiologies of different type of MPB become better understood, these distinguishing characteristics may shed additional light on potential mechanisms linking MPB and PC.

Although hormonal factors are thought to be possible causal factors of BPH, the association between MPB and BPH has been inconsistent. A case-control study found no association between baldness patterns and BPH [22]. Study conducted by Chen et al. [19] found the presence of baldness was an independent risk factor for the presence of BPH after adjusting for age, IPSS, urination time, urinary volume, abdominal obesity, glucose levels, insulin levels, and fibrinogen (OR = 5.14; 95% CI: 1.23–47.36; P = 0.041). Although our study has revealed a statistically significant association between MPB and BPH, further researches are necessary to better understand the potential association between BPH and MPB.

Several limitations of our meta-analysis merit discussion. First, our analysis includes 3 cohort studies and 14 case-control studies. Therefore, the confounding bias from included study per se cannot be excluded. Second, although we included only studies that used the Hamilton scale to evaluate baldness patterns, most studies included in this meta-analysis assessed baldness relied on self-reported which may introduce confounding factors.

5. Conclusions

We found baldness is positively associated with an increased risk of aggressive PC as well as BPH, compared to those without baldness. Despite our findings, further studies, preferably prospective cohort studies, are required to better elucidate these relationships and to advance knowledge in this field. In particular, the pathological mechanisms underlying our findings are far from clear and studies that are able to enhance our understanding in this area would be valuable.

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Authors' contributions

T.J. and T.W. have made substantial contributions to design the study. T.J. and J.M.L. have screened papers and conducted the quality rating and meta-analysis. The statistical analyses were conducted by S.D., X.D., and Y.T. . T.J. and T.W. have been involved in drafting the manuscript. All authors read and approved the final article.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at http://dx.doi.org/10.1016/j.urolonc.2017.09.022.

References

[1] Ellis JA, Sinclair R, Harrap SB. Androgenetic alopecia: pathogenesis and potential for therapy. Expert Rev Mol Med 2002;4:1–11, http://dx.doi.org/10.1017/S1462399402005112.
[2] VA: R. Androgens and human hair growth. Clin Endocrinol 1994;40:439–57.
[3] Black A, Pinsky PF, Grubb RL 3rd, Falk RT, Hsing AW, Chu L, et al. Sex steroid hormone metabolism in relation to risk of aggressive prostate cancer. Cancer Epidemiol Biomarkers Prev 2014;23:2374–82, http://dx.doi.org/10.1158/1055-9965.EPI-14-0700.
[4] Nicholson TM, Ricke WA. Androgens and estrogens in benign prostatic hyperplasia: past, present and future. Differentiation 2011;82:184–99, http://dx.doi.org/10.1016/j.diff.2011.04.006.
[5] Yassa M, Saliou M, De Rycke Y, Hemery C, Henni M, Bachaud JM, et al. Male pattern baldness and the risk of prostate cancer. Ann Oncol 2011;22:1824–7, http://dx.doi.org/10.1093/annonc/mdq695.
[6] Muller DC, Giles GG, Sinclair R, Hopper JL, English DR, Severi G. Age-dependent associations between androgenetic alopecia and prostate cancer risk. Cancer Epidemiol Biomarkers Prev 2013;22:209–15.
[7] Thomas JA, Antonelli Jodi A, Banez LL, Hoyo C, Grant D, Demark-Wahnefried W, et al. Androgenetic alopecia at various ages and prostate cancer risk in an equal-access multiethnic case-control series of veterans. Cancer Causes Control 2013;24:1045–52.
[8] Cremers RG, Aben KK, Vermeulen SH, den Heijer M, van Oort IM, Kiemeney LA. Androgenic alopecia is not useful as an indicator of men at high risk of prostate cancer. Eur J Cancer 46;3294–9.
[9] Wright JL, Page ST, Lin DW, Stanford JL. Male pattern baldness and prostate cancer risk in a population-based case-control study. Cancer Epidemiol 2010;34:131–5, http://dx.doi.org/10.1016/j.canep.2010.02.003.
[10] Amoretti A, Laydner H, Bergfeld W. Androgenetic alopecia and risk of prostate cancer: a systematic review and meta-analysis. J Am Acad Dermatol 2013;68:937–43, http://dx.doi.org/10.1016/j.jaad.2012.11.034.
[11] Downs SH, Black N. The feasibility of creating a checklist for the assessment of the methodological quality both of randomised and non-randomised studies of health care interventions. J Epidemiol Community Health 1998;52:377–84.
[12] Moher D, Liberati A, Tetzlaff J, Altman DG, Group P. Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement. PLoS Med 2009;6:e1000097, http://dx.doi.org/10.1371/journal.pmed1000097.
[13] Zeigler-Johnson C, Morales KH, Spangler E, Chang BL, Rebbeck TR. Relationship of early-onset baldness to prostate cancer in African-American men. Cancer Epidemiol Biomarkers Prev 2013;22:589–96, http://dx.doi.org/10.1158/1055-9965.EPI-12-0944.
[14] Giles GG, Severi G, Sinclair R, English DR, McCredie MR, Johnson W, et al. Androgenetic alopecia and prostate cancer: findings from an Australian case-control study. Cancer Epidemiol Biomarkers Prev 2002;11:549–53.
[15] Faydaci G, Bilal E, Necmettin P, Fatih T, Asuman O, Ugur K. Baldness, benign prostate hyperplasia, prostate cancer and androgen levels. Aging Male 2008;11:189–92, http://dx.doi.org/10.1080/13685530802400995.
[16] Denmark-Wahnefried W, Schildkraut JM, Thompson D, Lesko SM, McIntyre L, Schwingl P, et al. Early onset baldness and prostate cancer risk. Cancer Epidemiol Biomarkers Prev 2000;9:325–8.
[17] Demark-Wahnefried W, Lesko SM, Conaway MR, Robertson CN, Clark RV, Lobaugh B, et al. Serum androgens: associations with prostate cancer risk and hair patterning. J Androl 1997;18:495–500.
[18] Zhou CK, Pfeiffer RM, Cleary SD, Hoffman HJ, Levine PH, Chu LW, et al. Relationship between male pattern baldness and the risk of aggressive prostate cancer: an analysis of the prostate, lung, colorectal, and ovarian cancer screening trial. J Clin Oncol 2015;33:419–25, http://dx.doi.org/10.1200/JCO.2014.55.4279.
[19] Chen W, Yang C, Chen GY, Wu MC, Sheu HM, Tzai TS. Patients with a large prostate show a higher prevalence of androgenetic alopecia. Arch Dermatol Res 2004;296:245–9.
[20] Oh BR, Kim SJ, Moon JD, Kim HN, Kwon DD, Won YH, et al. Association of benign prostatic hyperplasia with male pattern baldness. Urology 1998;51:744–8.
[21] Barbosa JA, Muracca E, Nakano E, Paranhos M, Natalino R, Cordeiro P, et al. Risk factors for male lower urinary tract symptoms: the role of metabolic syndrome and androgenetic alopecia in a Latin American population. Urology 2013;82:182–8, http://dx.doi.org/10.1016/j.urology.2013.03.001.
[22] Fritschi L, Tabrizi J, Leavy J, Ambrosini G, Timperio A. Risk factors for surgically treated benign prostatic hyperplasia in Western Australia. Public Health 2007;121:781–9, http://dx.doi.org/10.1016/j.puhe.2007.01.011.
[23] Severi G, Sinclair R, Hopper JL, English DR, McCredie MR, Boyle P, et al. Androgenetic alopecia in men aged 40-69 years: prevalence and risk factors. Br J Dermatol 2003;149:1207–13.
[24] Sarre S, Määttänen L, Tammela TL, Auvinen A, Murtola TJ. Postscreening follow-up of the Finnish Prostate Cancer Screening Trial on putative prostate cancer risk factors: vitamin and mineral use, male pattern baldness, pubertal development and non-steroidal anti-inflammatory drug use. Scand J Urol 2016;50:267–73.
[25] Yip L, Rufaut N, Sinclairt R. Role of genetics and sex steroid hormones in male androgenetic alopecia and female pattern hair loss: an update of what we now know. Australas J Dermatol 2011;52:81–8.
[26] Zhu YS, Imperato-McGinley JL. 5 Alpha-reductase isozymes and androgen actions in the prostate. Ann N Y Acad Sci 2009;1155:43–56, http://dx.doi.org/10.1111/j.1749-6632.2009.04115.x.